EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 10

Basic Maths and Calculus

Differentiation Rules and Applications

By - Excellence Programme

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Topics to be covered

01 Differentiation: tangent slope at a point

02 Rate of change and the derivative

03 Quadratic equations and graph application

04 Basic differentiation formulae

05 Sum, difference, product and quotient rules

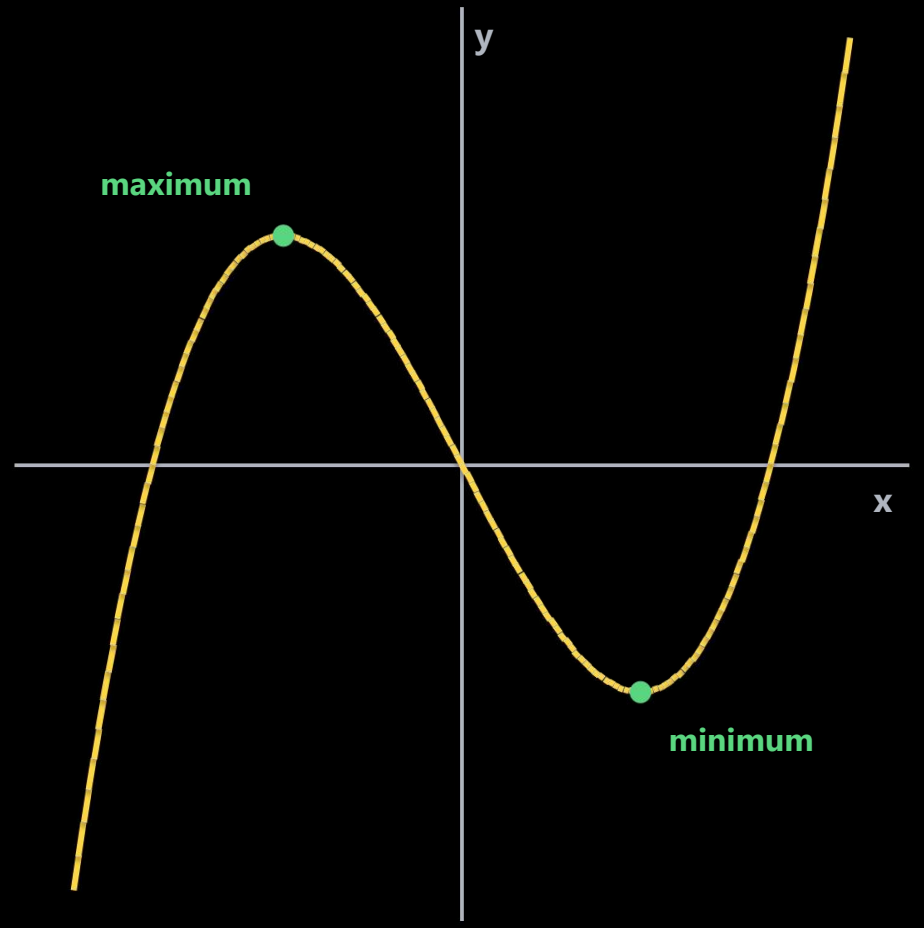
06 Mixed practice with simple solutions

Differentiation: slope at a point

Differentiation finds how quickly one quantity changes with another.

On a graph, it gives the slope of the tangent at the chosen point.

slope = Δy/Δx tangent slope = dy/dx



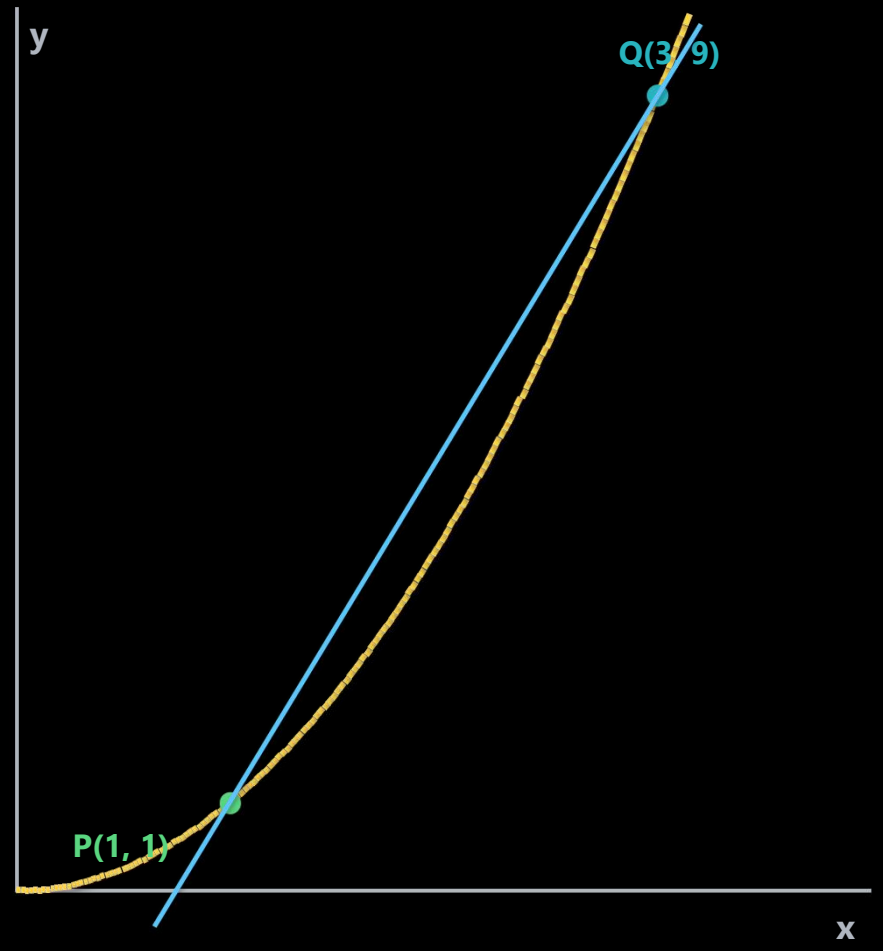
dy/dx = 0 at both turning points

Rate of change: from two points to one point

The average rate of change between two points is the secant's slope.

$$\text{average rate} = \Delta y/\Delta x = (y_2-y_1)/(x_2-x_1)$$

To find the rate at one point, bring the second point extremely close to the first.



The derivative is an instantaneous rate

Let x change by a small amount Δx. The corresponding change in y is Δy.

As Δx approaches zero, the secant approaches the tangent.

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

Therefore dy/dx gives the tangent's slope and the instantaneous rate of change.

Δx approaches zero; we do not divide by zero.

Rough work / extra work

Change in y for a linear relation

If $y = 2x+4$, find the change in y and the average rate of change when x changes from 1 to 3.

Solution: Change in y for a linear relation

Step 1: At $x_1 = 1$: $y_1 = 2(1)+4 = 6$

Step 2: At $x_2 = 3$: $y_2 = 2(3)+4 = 10$

Step 3: $\Delta y = y_2 - y_1 = 10-6 = 4$

Step 4: $\Delta y/\Delta x = 4/(3-1) = 2$

Change in y = 4; average rate of change = 2.

Increasing, decreasing and stationary

Graph behaviour	Derivative sign	Meaning
Increasing	$dy/dx > 0$	y rises as x increases
Stationary	$dy/dx = 0$	horizontal tangent
Decreasing	$dy/dx < 0$	y falls as x increases

The magnitude $|dy/dx|$ tells how steep the graph is.

Quadratic equation and its roots

A quadratic equation has the form $ax^2+bx+c = 0$, where $a \neq 0$.

$$x = (-b \pm \sqrt{(b^2-4ac)})/(2a)$$

sum of roots = -b/a

product of roots = c/a

Discriminant $D = b^2-4ac$	Nature of real roots
$D > 0$	two distinct real roots
$D = 0$	one repeated real root
$D < 0$	no real roots

Quadratic roots: $x^2-7x+12 = 0$

Find both roots by factorisation.

Solution: Quadratic roots: $x^2-7x+12 = 0$

Step 1: Find two numbers with product 12 and sum -7: -3 and -4

Step 2: $x^2-7x+12 = (x-3)(x-4)$

Step 3: $(x-3)(x-4) = 0$

$x = 3$ or $x = 4$.

Rough work / extra work

Quadratic roots: $x^2+7x-18 = 0$

Find both roots by factorisation.

Solution: Quadratic roots: $x^2+7x-18 = 0$

Step 1: Split 7x as 9x-2x

Step 2: $x^2+9x-2x-18 = 0$

Step 3: $(x+9)(x-2) = 0$

x = -9 or x = 2.

Quadratic roots: $x^2+8x+15 = 0$

Find both roots by factorisation.

Solution: Quadratic roots: $x^2+8x+15 = 0$

Step 1: Find two numbers with product 15 and sum 8: 5 and 3

Step 2: $x^2+8x+15 = (x+5)(x+3)$

Step 3: $(x+5)(x+3) = 0$

$x = -5$ or $x = -3$.

Rough work / extra work

Parabola $y = x^2-3x-10$

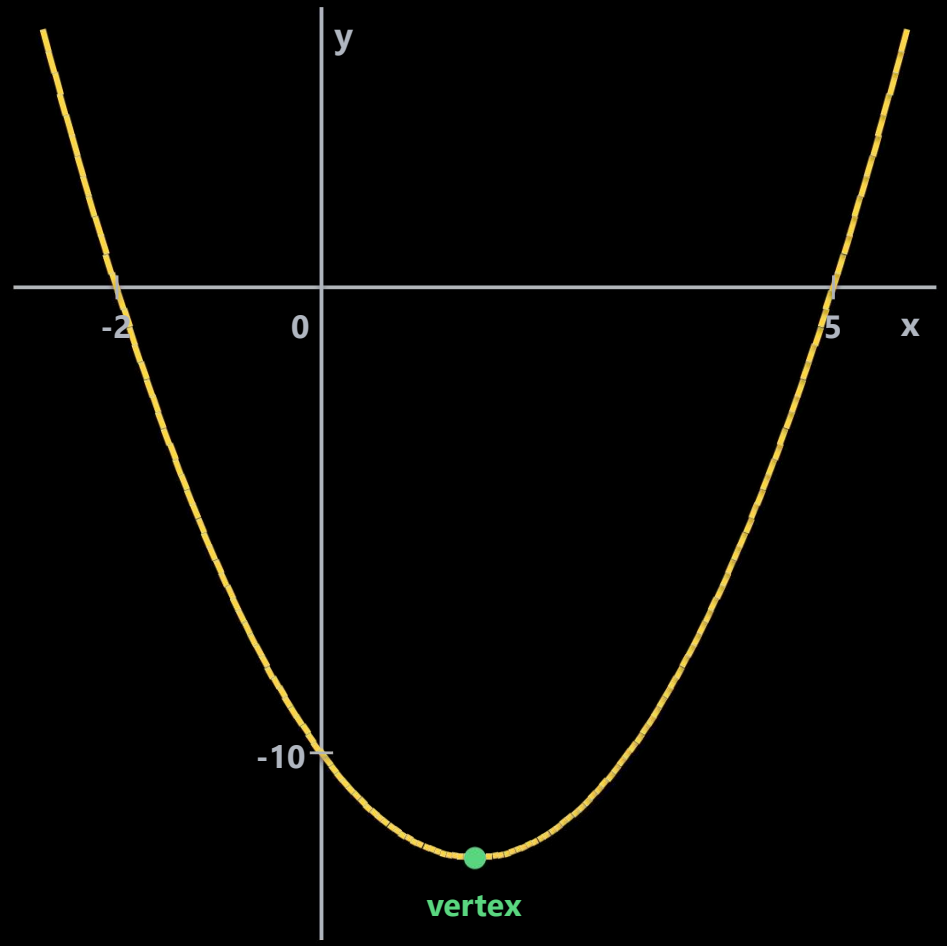
x-intercepts: solve $x^2-3x-10 = (x-5)(x+2) = 0$.

roots: x = -2 and x = 5

y-intercept: (0, -10)

turning point: x = 3/2

At x = 3/2, y = -49/4 = -12.25.



Physics application: downward motion

A ball is thrown vertically downward at 10 m s^{-1} from a height of 75 m. Taking g = 10 m s^{-2}, find its time of flight.

Solution: Physics application: downward motion

Step 1: Choose downward as positive: $u = 10$, $a = 10$, $s = 75$

Step 2: Use $s = ut+1/2\ at^2$

Step 3: $75 = 10t+5t^2$, so $t^2+2t-15 = 0$

Step 4: $(t+5)(t-3) = 0$

Time cannot be negative, so $t = 3$ s.

Average rate becomes a derivative

For y = 3x-2:

y(1) = 1 and y(3) = 7, so Δy = 6

Δy/Δx = 6/2 = 3

Because the graph is a straight line, the rate is 3 everywhere.

dy/dx = d(3x-2)/dx = 3

Read the differential operator correctly

d/dx means: differentiate the expression that follows with respect to x.

d(y)/dx: rate of change of y with respect to x

d(P)/dx: rate of change of P with respect to x

d(x)/dt: rate of change of x with respect to time t

Always identify both the changing quantity and the independent variable.

Rough work / extra work

Derivative of a constant

If $y = C$, then $dy/dx = 0$

A constant does not change when x changes, so its rate of change is zero.

$d(6)/dx = 0$

$d(e^0)/dx = d(1)/dx = 0$

$d(\sin^2\theta+\cos^2\theta)/d\theta = d(1)/d\theta = 0$

Power rule for algebraic functions

For constant n: $d(x^n)/dx = n\,x^{n-1}$

Multiply by the old power, then reduce the power by one.

$d(x^2)/dx = 2x$

$d(x^3)/dx = 3x^2$

$d(1/x^2)/dx = d(x^{-2})/dx = -2x^{-3}$

Exponential and trigonometric derivatives

Function	Derivative
e^x	e^x
sin x	cos x
cos x	-sin x
tan x	$\sec^2 x$
$\log_e x$	1/x

Angles must be in radians when using these standard trigonometric derivatives.

Constant multiple rule

$$d[k\,f(x)]/dx = k\,f'(x)$$

A constant multiplier stays outside differentiation.

If $y = 3x^5$, then $dy/dx = 3(5x^4) = 15x^4$

If $y = 7\sin x$, then $dy/dx = 7\cos x$

Sum and difference rules

$$d(A \pm B)/dx = dA/dx \pm dB/dx$$

Differentiate every term separately and keep its sign.

$d(\sin x + x^2)/dx = \cos x + 2x$

$d(e^x - \cos x)/dx = e^x + \sin x$

Rough work / extra work

Product and quotient rules

PRODUCT

$$d(AB)/dx = A\,dB/dx + B\,dA/dx$$

Differentiate one factor at a time.

QUOTIENT

$$d(A/B)/dx = (B\,dA/dx - A\,dB/dx)/B^2$$

Keep the denominator squared.

For a quotient, remember: denominator × derivative of numerator minus numerator × derivative of denominator.

Differentiate a simple polynomial

For $y = 4x^3-5x^2+2x-7$, find dy/dx.

Solution: Differentiate a simple polynomial

Step 1: $d(4x^3)/dx = 12x^2$

Step 2: $d(-5x^2)/dx = -10x$

Step 3: $d(2x)/dx = 2$ and $d(-7)/dx = 0$

$dy/dx = 12x^2-10x+2.$

Product-rule practice

Differentiate $y = \tan x \cdot \log_e x$.

Solution: Product-rule practice

Step 1: Use $d(AB)/dx = A\,dB/dx + B\,dA/dx$

Step 2: $d(\tan x)/dx = \sec^2 x$ and $d(\log_e x)/dx = 1/x$

$dy/dx = \tan x/x + \log_e x \cdot \sec^2 x$.

Rough work / extra work

Stationary-point practice

If $y = A/x^3 - B/x^4$, find x for which $dy/dx = 0$. Assume A and B are non-zero constants.

Solution: Stationary-point practice

Step 1: Write $y = Ax^{-3} - Bx^{-4}$

Step 2: $dy/dx = -3Ax^{-4} + 4Bx^{-5}$

Step 3: Set zero and multiply by x^5: $-3Ax + 4B = 0$

$x = 4B/(3A)$.

Quotient-rule practice: exponential

Differentiate $y = e^x/x^2$, where $x \neq 0$.

Solution: Quotient-rule practice: exponential

Step 1: Take $A = e^x$ and $B = x^2$

Step 2: $dy/dx = [x^2e^x - e^x(2x)]/x^4$

Step 3: Factor e^x x from the numerator

$dy/dx = e^x(x-2)/x^3$.

Quotient-rule practice: algebraic

Differentiate $y = (x^2-1)/(x^2+1)$.

Solution: Quotient-rule practice: algebraic

Step 1: Use the quotient rule with $A = x^2-1$ and $B = x^2+1$

Step 2: $dy/dx = [(x^2+1)(2x)-(x^2-1)(2x)]/(x^2+1)^2$

Step 3: The numerator becomes $2x[(x^2+1)-(x^2-1)] = 4x$

$dy/dx = 4x/(x^2+1)^2$.

Rough work / extra work

Mixed trigonometric-algebraic practice

Differentiate $y = \cos x + 2/x^3$, where $x \neq 0$.

Solution: Mixed trigonometric-algebraic practice

Step 1: Write $2/x^3$ as $2x^{-3}$

Step 2: $d(\cos x)/dx = -\sin x$

Step 3: $d(2x^{-3})/dx = -6x^{-4}$

$dy/dx = -\sin x - 6/x^4$.

Simplify before differentiating

Differentiate $y = (\sqrt{x} + 1/\sqrt{x})^2$ for $x > 0$.

Solution: Simplify before differentiating

Step 1: Expand first: $y = x+2+1/x$

Step 2: Write $1/x$ as x^{-1}

Step 3: $dy/dx = 1+0-x^{-2}$

$dy/dx = 1-1/x^2$.

Rough work / extra work

Differentiation lock lines

dy/dx is the instantaneous rate of change and tangent slope

$d(C)/dx = 0$ $d(x^n)/dx = nx^{n-1}$

differentiate sums term by term; constants stay outside

product: $A B' + B A'$ quotient: $(B A' - A B')/B^2$

At a smooth maximum or minimum, dy/dx = 0.

Practice

1. Redo every question without looking at its solution.

2. Write the six basic derivative formulae from memory.

3. Draw one curve and label positive, zero and negative slope regions.

4. Check signs carefully in product and quotient questions.

Always simplify the function first when it makes differentiation shorter.